

Adversarial Attacks

Robust Deep Learning (Seminar)

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Which one is the gibbon?



Which one is the gibbon?



panda



gibbon

A fragile foundation



x

“panda”

57.7% confidence

$+ .007 \times$



$\text{sign}(\nabla_x J(\theta, x, y))$

“nematode”

8.2% confidence

$=$



$x +$

$\epsilon \text{sign}(\nabla_x J(\theta, x, y))$

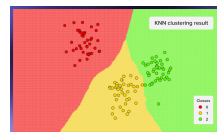
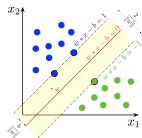
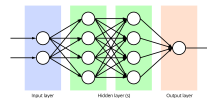
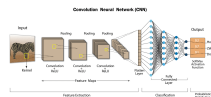
“gibbon”

99.3 % confidence

Goodfellow et al., 2014

Transferability

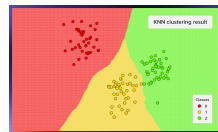
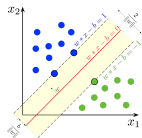
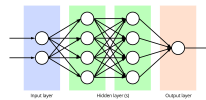
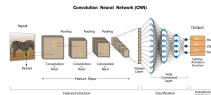
- An adversarial example crafted against one model often fools other models as well



Papernot et al., 2016

Transferability

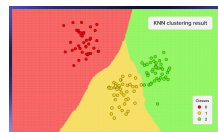
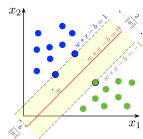
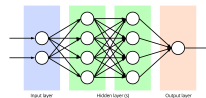
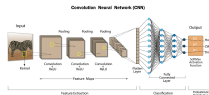
- ▶ An adversarial example crafted against one model often fools other models as well
- ▶ Survives a change of architecture as well as of training data, initialisation, hyperparameters



Papernot et al., 2016

Transferability

- ▶ An adversarial example crafted against one model often fools other models as well
- ▶ Survives a change of architecture as well as of training data, initialisation, hyperparameters
- ▶ Vulnerability apparently inherent to *problem itself*, not to one model or one architecture



Papernot et al., 2016

Why this matters

- **Fundamental blind spot:** model and human do not share one understanding of input



Images: medicalimaging.na; evsmarts.com; thetravelimages.com

Why this matters

- ▶ **Fundamental blind spot:** model and human do not share one understanding of input
- ▶ Raises questions about **reliability** of systems



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Why this matters

- ▶ **Fundamental blind spot:** model and human do not share one understanding of input
- ▶ Raises questions about **reliability** of systems
- ▶ More pressing as systems are entering deeper and deeper into **everyday life** and critical infrastructure



Images: medicalimaging.na; evsmarts.com; thetravelimages.com

Outline

1. What is an adversarial example?
2. Taxonomy
3. Evasion attacks
4. Poisoning attacks
5. Discussion
 - 5.1 Physical adversarial attacks
 - 5.2 Real-life adversarial attacks
 - 5.3 Explaining adversarial examples
 - 5.4 Adversarial attacks as a subject of study

Outline

1. What is an adversarial example?
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Definition

Definition 1 (Adversarial Example)

An *adversarial example* is an **input to a machine learning model** that has been **intentionally perturbed**, typically **in a subtle and often imperceptible manner**, such that the model produces an incorrect prediction while the semantic content of the input remains unchanged.

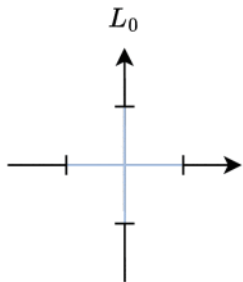
Definition

Definition 1 (Adversarial Example)

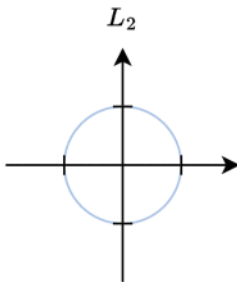
An *adversarial example* is an **input to a machine learning model** that has been **intentionally perturbed**, typically **in a subtle and often imperceptible manner**, such that the model produces an incorrect prediction while the semantic content of the input remains unchanged.

- ▶ **Machine learning models** — no quirk of deep networks, but a general property of machine learners.
- ▶ **Intentional perturbations** — generated systematically through optimisation and exploited by an adversary, not stumbled upon.
- ▶ **Subtle and imperceptible perturbations** — mathematically small under a chosen norm, and visually indistinguishable to humans; the latter is asserted far more often than it is rigorously validated.

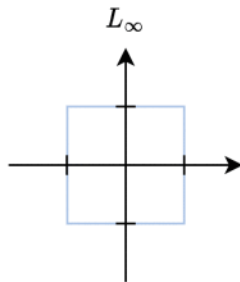
Measuring “small”: the ℓ_p norms



$$\|x\|_0 = (|x_1|^0 + \dots + |x_n|^0)$$



$$\|x\|_2 = (|x_1|^2 + \dots + |x_n|^2)^{\frac{1}{2}}$$



$$\|x\|_\infty = \max_i |x_i|$$

The norm defines the **perturbation budget** ϵ and, with it, what “similar” even means.

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Four orthogonal axes

- ▶ **Attacker knowledge** — white-box vs. black-box
- ▶ **Attack stage** — evasion (test time) vs. poisoning (train time)
- ▶ **Attack goal** — targeted vs. untargeted
- ▶ **Perturbation constraint** — which norm, which budget

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$$\min_{\delta} \|\delta\|_2 \quad \text{subject to} \quad f(x + \delta) = t$$

- ▶ f has no gradient — it returns a discrete class label, $f(x + \delta) = t$ is a hard, non-differentiable condition.
 - ▶ Derivative-based optimisers such as L-BFGS need a gradient to move at all.
- ↪ Reformulate.

$$\min_{\delta} c \|\delta\|_2 + L_f(x + \delta, t)$$

Significance

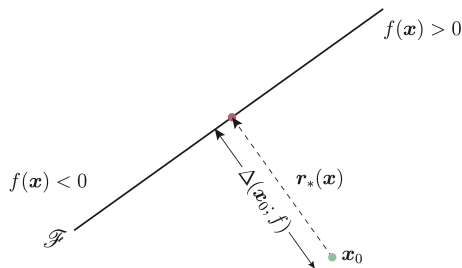
- ▶ First method to **reliably generate** adversarial examples.
- ▶ Computationally expensive — far too slow to be practical.
- ↪ It *proved a point* rather than providing a usable attack.

The arc of the field

2013 **L-BFGS** — expensive, but proves concept

- ▶ For an affine $f(x) = w^\top x + b$, the decision boundary $\mathcal{F}$ is a **hyperplane**.
- ▶ Minimal perturbation is the **orthogonal projection** of x_0 onto $\mathcal{F}$ — closed form, no search.

$$r_*(x_0) = -\frac{f(x_0)}{\|w\|_2^2} w$$

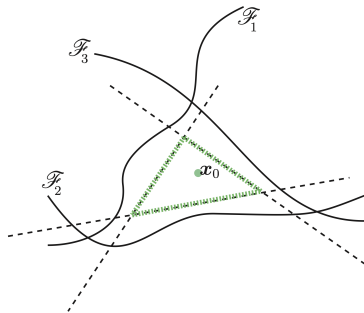


DeepFool

The general, multiclass case

White-box · Evasion · Untargeted · ℓ_2

- ▶ **Non-linear:** iterative $\rightarrow$ linearise the score function at current point (first-order Taylor), project onto that hyperplane, repeat until label flips.
- ▶ **Multiclass:** Linearise every class boundary, project onto the **closest**.



Significance

- ▶ Gave problem a **geometric reading**: robustness as distance to the decision boundary.
- ▶ First to **extend to any ℓ_p norm**, $p \in [1, \infty)$.

The arc of the field

2013 **L-BFGS** — expensive, but proves concept

2016 **DeepFool** — minimal perturbation via iterative linear approximation

$$\underbrace{\min_{\delta} c \|\delta\|_2 + L_f(x + \delta, t)}_{\text{L-BFGS}} \longrightarrow \underbrace{\min_{\delta} \|\delta\|_p + c \cdot \ell(x + \delta)}_{\text{C\&W}}$$

- ▶ $\ell(x') = \max(\max_{i \neq t} z_i(x') - z_t(x'), 0)$.
- ▶ Built so that $\ell(x') \leq 0 \iff$ input is classified as $t \rightarrow$ faithfully encodes the attack goal.
- ▶ c decided through **binary search**

$$x' = \frac{1}{2}(\tanh(w) + 1), \quad w \in \mathbb{R}^m$$

- ▶ mapping any $w \in \mathbb{R}^m$ into $(0, 1)^m \rightarrow$ every w decodes to a **valid image** $\rightarrow$ dissolving box-constraint
- ▶ Optimise w **unconstrained** with Adam optimiser

Significance

- ▶ Strongest attack of its time across $\ell_0/\ell_2/\ell_\infty$.
- ▶ Broke *defensive distillation*, exposing it as *gradient masking*

The arc of the field

2013 **L-BFGS** — expensive, but proves concept

2016 **DeepFool** — minimal perturbation via iterative linear approximation

2017 **C&W** — reformulate optimisation

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Physical adversarial attacks



Sharif et al., 2016 · Xu et al., 2020



Eykholt et al., 2018



Athalye et al., 2018

Real-life adversarial attacks

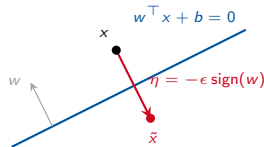
- ▶ Confirmed malicious attacks on real institutions or individuals — strikingly rare
- ▶ Two candidate explanations:
 1. they slip through unnoticed, logged as ordinary misclassifications
 2. they are simply of little use to actual attackers
- ▶ Case for the second:
 - ▶ crafting them reliably needs ML *and* security expertise
 - ▶ DDoS, character obfuscation, packing a binary — cheap, fast, effective
 - ~> picking a lock where a crowbar is cheaper and works just as well

Explaining adversarial examples

High-dimensional linearity

Goodfellow et al., 2014

- ▶ Affine case first: perturbation shifts activation by $w^\top \eta$
 - ▶ Worst case: η aligned with the weights
 - ▶ Output change — **linear** in n (number of dimensions)
- ~> Many imperceptible, weight-aligned changes accumulate into one large shift



$$\begin{aligned} w^\top \tilde{x} &= w^\top (x + \eta) \\ &= w^\top x + w^\top \eta \end{aligned}$$

$$w^\top \eta = \epsilon \sum_{i=1}^n |w_i| = \epsilon m n$$

n dimensions, mean weight magnitude m

From linear models to deep networks

Goodfellow et al., 2014

- ▶ Modern architectures deliberately behave *locally linearly* (easier to optimise)
- ↪ The linear analysis carries over within ϵ -ball

ReLU

sigmoid near zero

LSTM

maxout

linear-softmax

Predictive vs. robust features

Ilyas et al., 2019

Predictive

Correlated with the label in expectation.

Robust

Stays correlated *under* adversarial perturbation.

→ **Predictive but non-robust:** Standard training happily relies on such features.

Non-robust features are real signal

Ilyas et al., 2019



Why that works

Ilyas et al., 2019

	Robust features point at	Non-robust features point at	Label
Original clean training image (in $\mathcal{D}$)	dog	dog	dog
Perturbed training image (in $\hat{\mathcal{D}}_{NR}$)	dog <i>(unaffected by tiny perturbation)</i>	cat <i>(through adversarial perturbation)</i>	cat
Original clean test image	cat	cat	cat (predicted)

Adversarial attacks as a subject of study

- ▶ Not really significant security threat
- ▶ Mainly *epistemic significance*: not about what attacker can do but more about what they reveal about the systems they target
- ▶ Before them: accuracy was almost only metric, robustness assumed implicitly
- ▶ After them:
 - ▶ high test accuracy $\neq$ grasping the task as a human would
 - ▶ a more intricate account of how learners operate
 - ▶ robustness as a distinct property of interest

FINAL SLIDE?

References I